



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ADAPTIVE SOFTWARE ENGINEERING – (24CI3201)

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TITLE OF THE PROJECT		
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Abstract

Modern software applications increasingly rely on cloud-native architectures, microservices, containers, and continuous software delivery. While DevOps and DevSecOps practices have significantly improved the speed of software development and deployment, managing distributed applications has become increasingly complex. Development teams must continuously monitor application health, analyze system failures, identify security vulnerabilities, manage deployments, and respond to operational incidents. Traditional approaches often depend on manual monitoring and reactive troubleshooting, which can increase downtime, delay deployments, and make it difficult to identify potential failures before they occur. IntelliOps AI: Autonomous DevSecOps Platform for Predictive Software Delivery is proposed as an intelligent platform that integrates DevOps, DevSecOps, cloud-native technologies, monitoring, and Artificial Intelligence to provide a unified environment for secure, reliable, and predictive software delivery.

The proposed platform is designed to automate major stages of the software delivery lifecycle, beginning with source-code management and continuing through continuous integration, security validation, containerization, deployment, monitoring, and automated recovery. Developers can manage their projects through a centralized dashboard and use Git-based workflows to trigger automated CI/CD pipelines. The pipeline performs application building and testing, static code analysis, and container vulnerability scanning before deployment. Tools such as GitHub Actions, SonarQube, Trivy, Docker, and Kubernetes can be integrated to establish a secure and automated software delivery process.

A major component of IntelliOps AI is its AI-driven predictive intelligence layer. Instead of relying only on threshold-based monitoring, the platform collects operational metrics such as CPU utilization, memory consumption, request rate, response time, error rate, and service health. These metrics are processed by a machine learning model to identify abnormal behavior and predict potential service failures. Based on historical and real-time data, the system can classify the health or risk level of a service as healthy, warning, or critical. The platform can also calculate a deployment-readiness or deployment-risk score by considering factors such as test results, code quality, security vulnerabilities, previous deployment failures, and system performance.

The platform further introduces an automated self-healing mechanism for handling selected operational failures. When a service becomes unhealthy or the predictive model identifies a high-risk condition, the system can initiate predefined recovery actions such as restarting a failed container, restarting a Kubernetes pod, or recommending resource scaling. This changes the operational approach from reactive incident handling to proactive failure prevention and automated recovery. All actions performed by the system are recorded in centralized logs and presented through the dashboard for transparency and auditing.

For observability, IntelliOps AI integrates Prometheus, Grafana, and application health metrics to provide real-time visibility into microservice performance. Administrators and developers can monitor service availability, CPU and memory utilization, request rates, response times, error rates, deployment status, and recovery events. The security layer incorporates DevSecOps practices such as static code analysis, dependency and container vulnerability scanning, and dynamic application security testing. These mechanisms help identify security weaknesses earlier in the software development lifecycle rather than after deployment.

The proposed system can be implemented using a modern technology stack consisting of React.js for the web dashboard, Spring Boot and Spring Cloud for backend microservices and API management, PostgreSQL for persistent data storage, Docker for containerization, Kubernetes for orchestration, GitHub Actions for CI/CD automation, SonarQube and Trivy for security and quality analysis, and Prometheus and Grafana for monitoring and visualization. The AI component can be developed using Python, Pandas, Scikit-learn, and MLflow, enabling model training, evaluation, versioning, and deployment. The platform can initially be deployed locally using Docker Compose and Minikube and can later be extended to a cloud environment such as AWS.

The overall objective of IntelliOps AI is to create a unified, intelligent, and partially autonomous software delivery environment that reduces manual operational effort while improving application reliability, security,

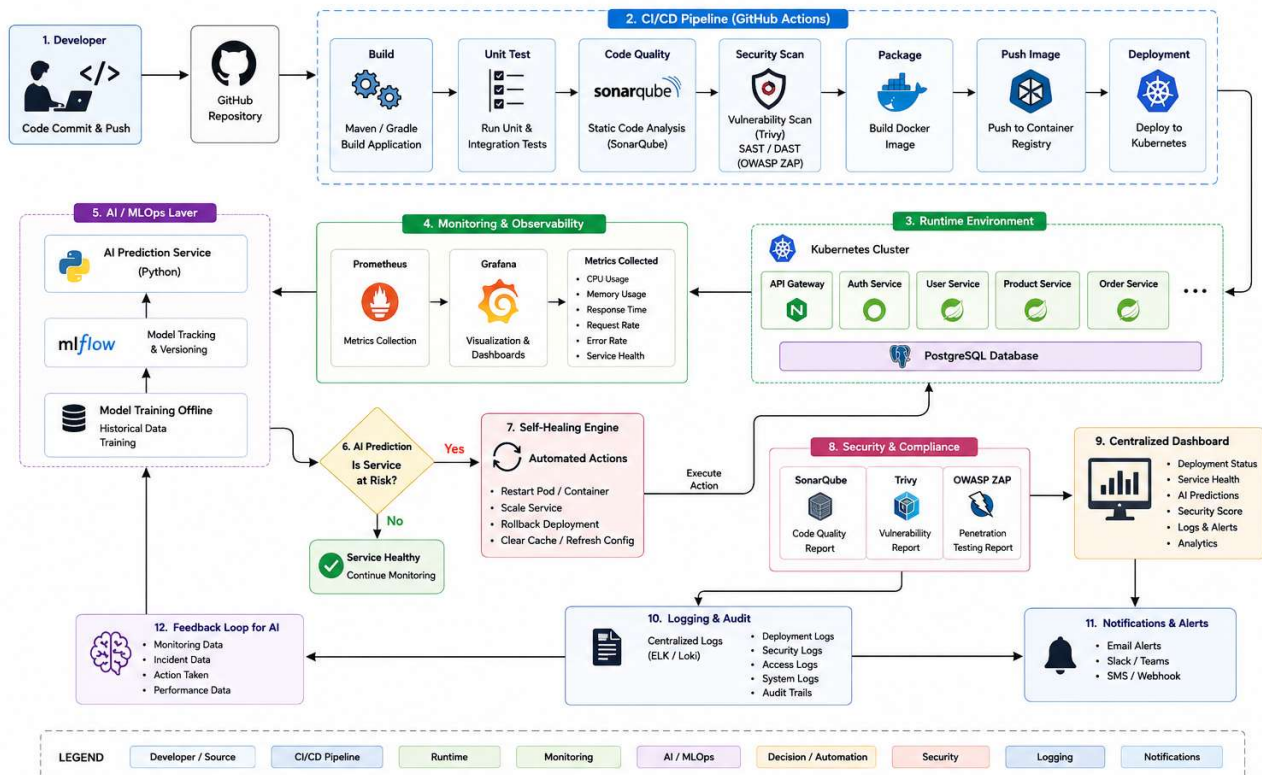
and deployment confidence. The project demonstrates how Agile software engineering, DevOps, DevSecOps, cloud-native computing, monitoring, and MLOps can be integrated into a single practical system. By combining predictive analytics with automated deployment and recovery mechanisms, the proposed platform aims to enable faster and safer software delivery while providing developers with actionable insights into application health and operational risks.

Keywords: DevSecOps, DevOps, Artificial Intelligence, Machine Learning, Self-Healing, Microservices, Predictive Analytics, CI/CD, Kubernetes, Docker, Cloud-Native Computing, Observability, MLOps, Automated Deployment.

Work Flow

IntelliOps AI: Autonomous DevSecOps Platform for Predictive Software Delivery

Workflow Diagram



Signature of the Guide

HOD